

Microeconomics II



What is it?

Topic 8: Price discrimination

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Introduction and roadmap

- **Price discrimination** = selling different units of output at different prices
- **Roadmap**
 - 1st degree price discrimination
 - 2nd degree price discrimination
 - 3rd degree price discrimination

	1st degree <i>„perfect discrimination“</i>	2nd degree <i>„non-linear pricing“</i>	3rd degree <i>„market segmentation“</i>
Different price for different units?	Yes	Yes	No
Different price for different buyers?	Yes	No	Yes

- Other pricing strategies
 - Bundling
 - Tie-ins

Motivation

- Understand pricing strategies by real-world companies



- Important implications for regulation and anti-trust



1st degree price discrimination

- „*Perfect discrimination*“

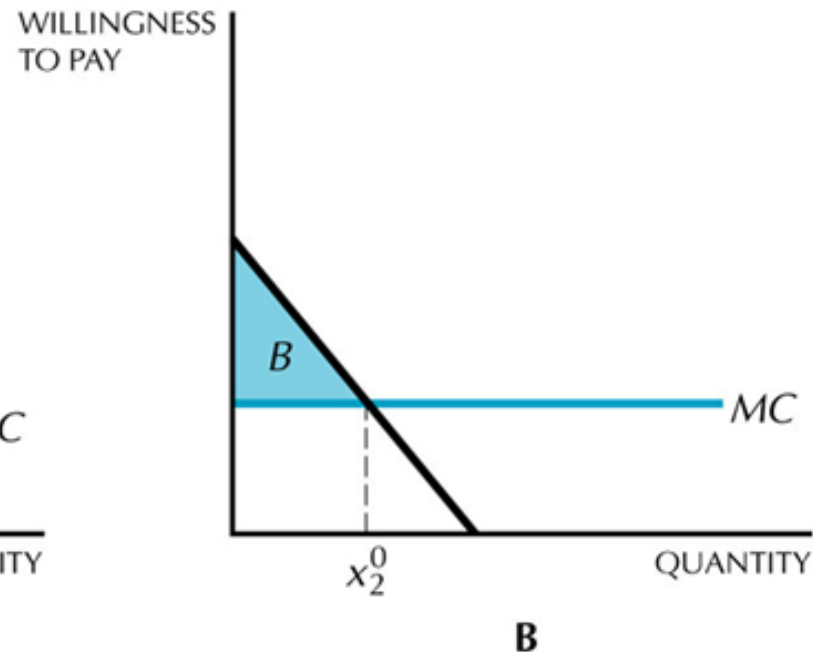
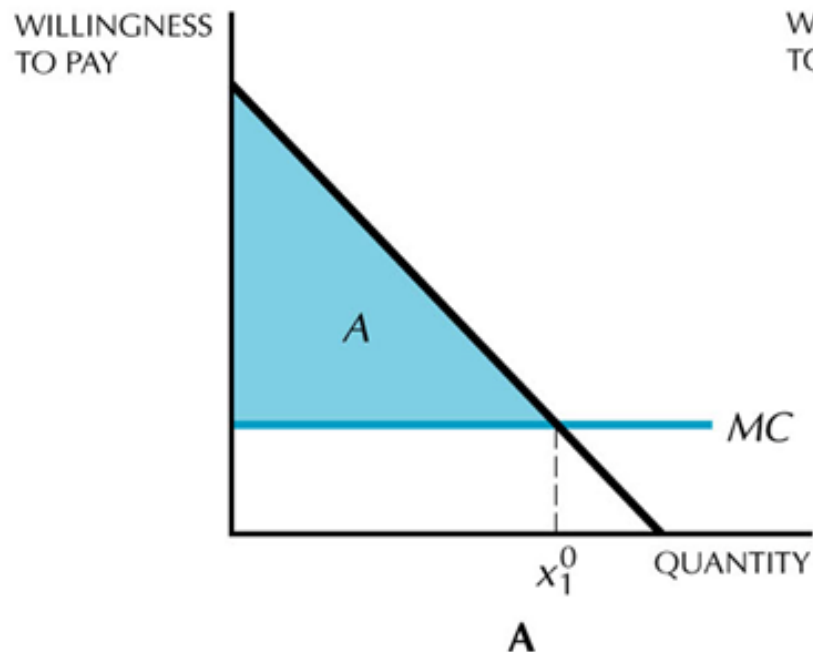
Definition

- Price different for **different individuals** and **different units**

1st degree price discrimination

Analysis

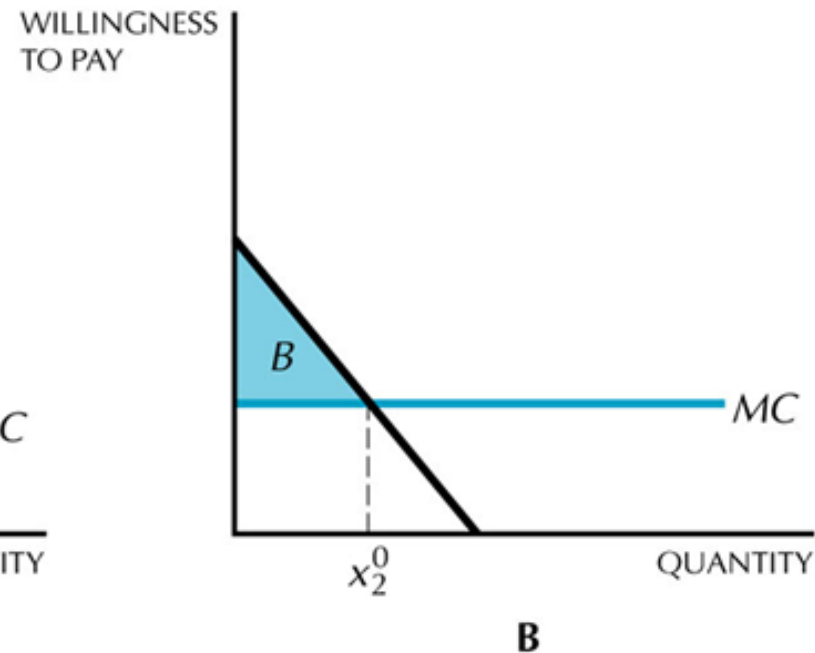
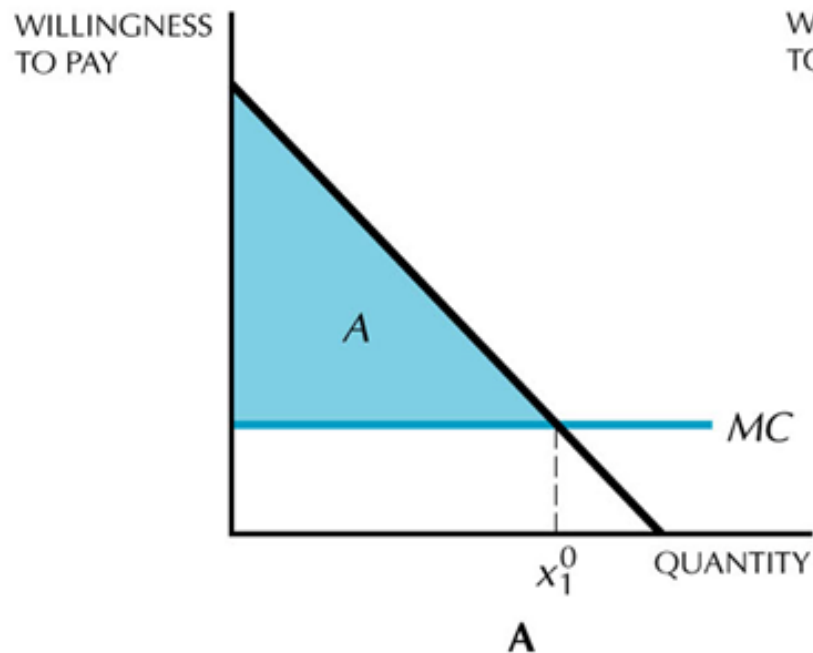
- Each unit sold at each consumer's reservation price.
- Similar to selling a fixed amount of good at "take it or leave it" price.
- For the last sold unit $P = MC \rightarrow$ Efficient (like perfect competition)
- No consumer surplus $\rightarrow$ All surplus goes to the seller (very much unlike perfect competition!)



1st degree price discrimination

Key assumptions

- Arbitrage not possible
- Seller knows each consumer's demand schedule



1st degree price discrimination

“Perfect discrimination” rare in practice but some examples are...

- University tuition in the US



?

- Accountant, lawyer, consultant, therapist...



- Proprietary data

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2nd degree price discrimination

- „*Non-linear pricing*“ or „*menu pricing*“

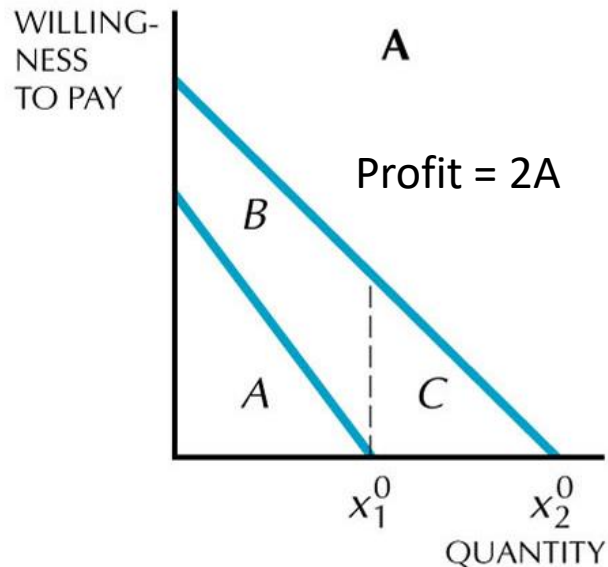
Definition

- Price different for **different units** but the same across individuals
- The key point is consumers self-select into the quantity or the specification

2nd degree price discrimination

Analysis

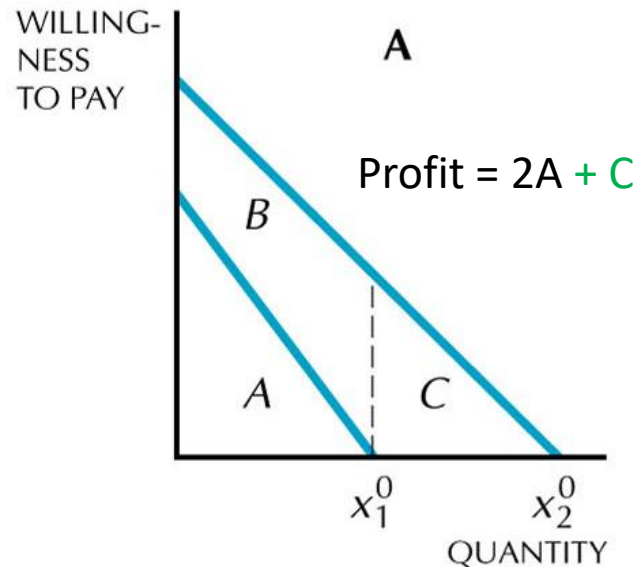
- Assume: $MC=0$, consumers 1 (low willingness to pay, WTP) and 2 (high WTP)
- The seller would like to sell
 - to consumer 1: quantity x_1 for price A
 - to consumer 2: quantity x_2 for price $A+B+C$
- ...but consumer 2 has an incentive to **pretend to be consumer 1**



2nd degree price discrimination

Analysis

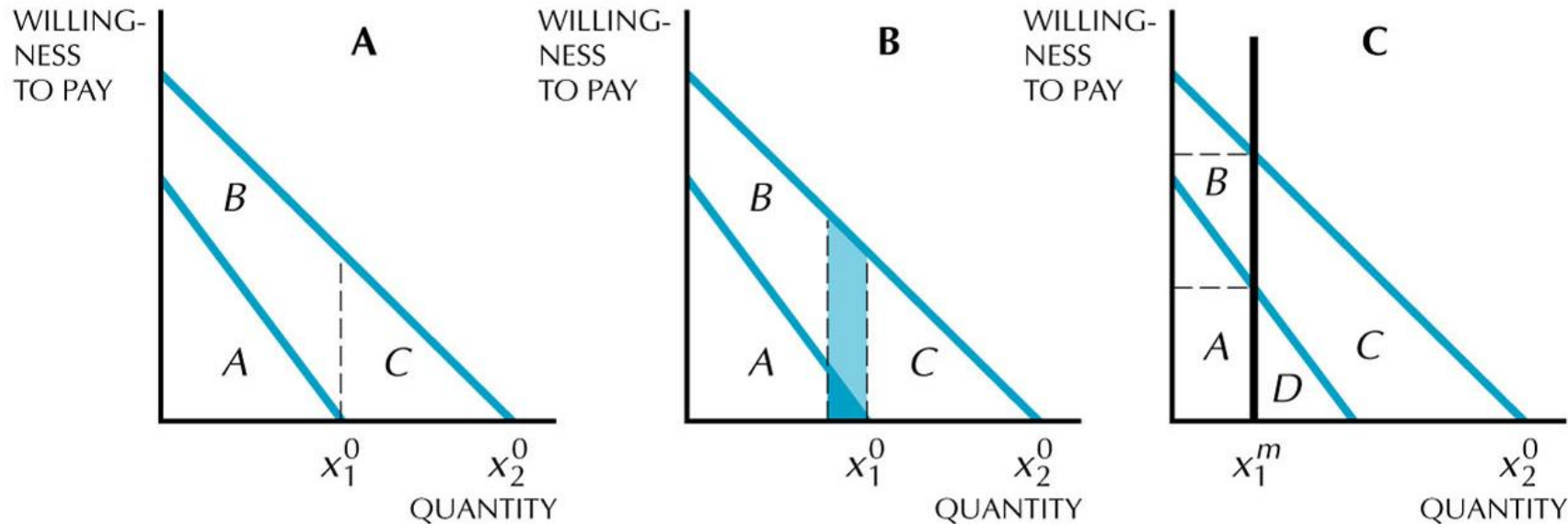
- The seller offers a **menu**: charge A for x_1 and $A+C$ for x_2
→ consumers self-select into x_1 and x_2
- But the seller can further increase profits by reducing x_1



2nd degree price discrimination

Analysis

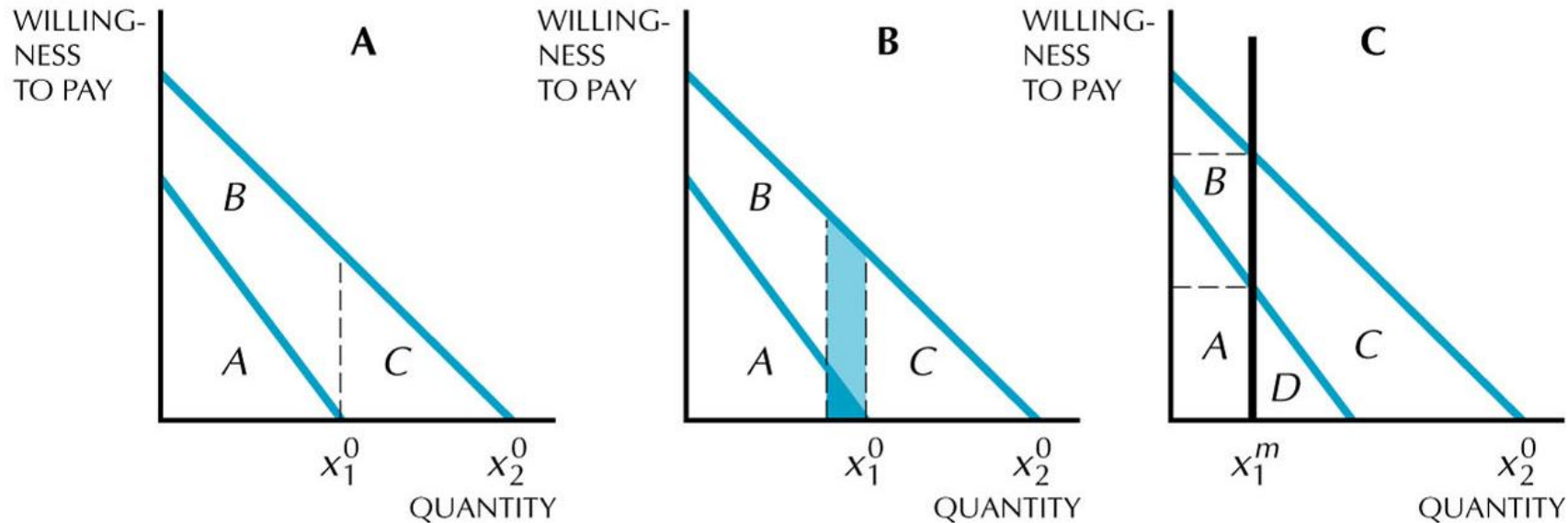
- The seller offers a **menu**: charge A for x_1 and $A+C$ for x_2
→ consumers self-select into x_1 and x_2
- But the seller can further increase profits by reducing x_1
- Optimum: the marginal reduction in profit on consumer 1 equals the marginal increase in profit on consumer 2



2nd degree price discrimination

Implications

- Consumer 1 (low WTP) has no surplus
- Consumer 2 (low WTP) has a strictly positive surplus = B
- For the last unit sold to consumer 2, $P = MC \rightarrow$ Efficient
- For the last unit sold to consumer 1, $P > MC \rightarrow$ Inefficient



2nd degree price discrimination

Examples

- Bulk discounts



- Cell-phone plans



- Discount cards (IN25/50)



2nd degree price discrimination

The quality interpretation

- Products with different features rather than different quantity
- Cheaper versions sometimes made intentionally worse, even at extra costs

- **Examples:**

- Electronics

Enthusiast camera



Professional camera



- Flights

- Eg. Saturday night stayover, luggage limits, boarding

- Premium features



3rd degree price discrimination

„Market segmentation“

Definition

- Price different for **different individuals** but the same across units

Assumptions:

- Seller able to identify the different (groups of) individuals
- Seller able to prevent arbitrage

3rd degree price discrimination

Analysis

- The seller maximises $p_1(x_1)x_1 + p_2(x_2)x_2 - c(x_1 + x_2)$
- In the optimum:
 - $MR_1(x_1) = MR_2(x_2) = MC(x_1 + x_2)$
 - $p_1(x_1) \left[1 - \frac{1}{|\varepsilon_1(x_1)|}\right] = p_2(x_2) \left[1 - \frac{1}{|\varepsilon_2(x_2)|}\right] = MC(x_1 + x_2)$
 - If $p_1 > p_2, |\varepsilon_1(x_1)| < |\varepsilon_2(x_2)|$
- The seller charges **higher price in the less price-elastic** market segment

3rd degree price discrimination

Examples

- Student discounts



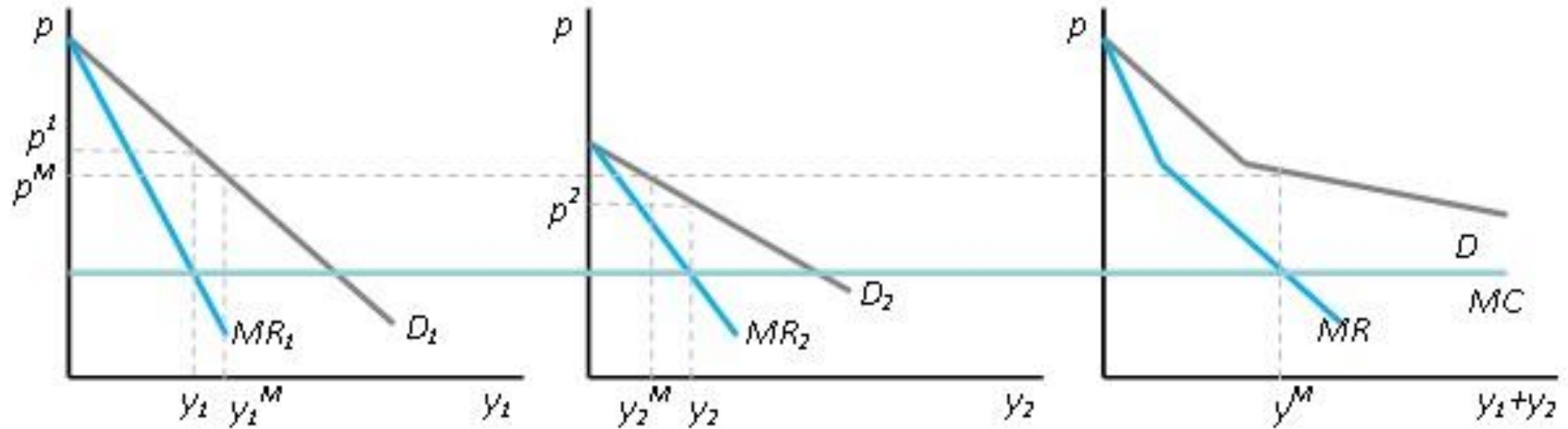
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- Pharmaceuticals



3rd degree price discrimination

Price setting with and without discrimination



3rd degree price discrimination

Are all markets served without discrimination?

- Setup

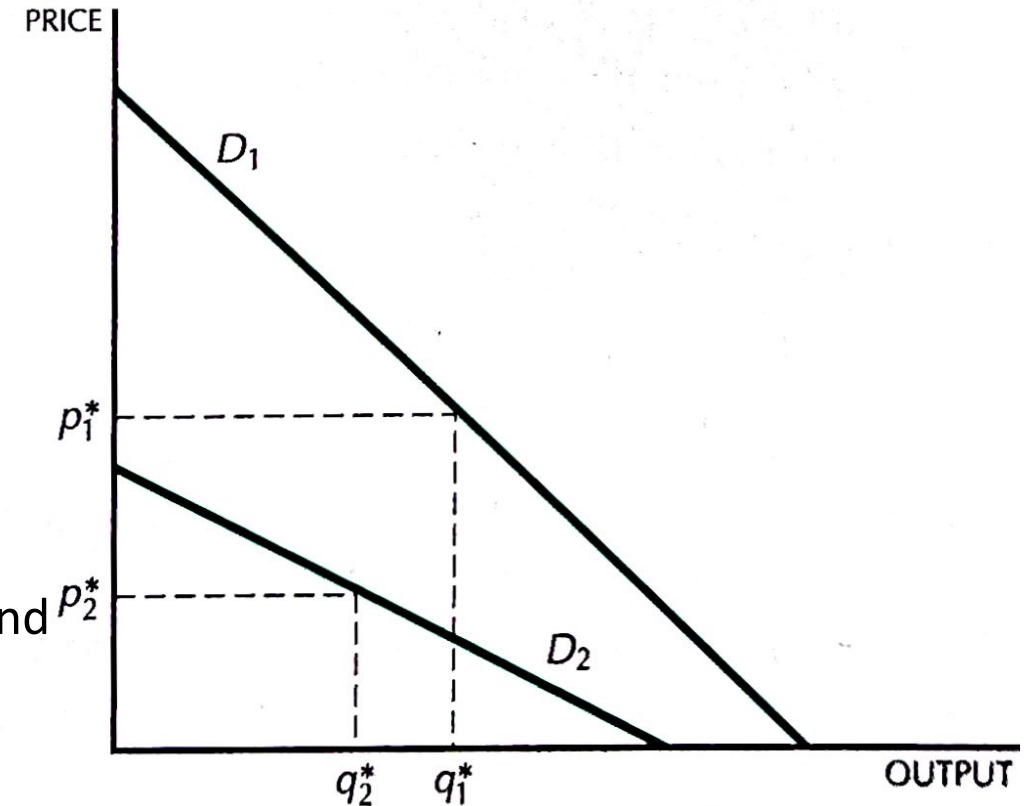
- Market 1 (high demand): $x_1 = a - bp_1$
- Market 2 (low demand): $x_2 = c - dp_2$
- $MC = 0$

- If selling in both markets w/o discrimination

- Discrimination: $x_1^* = a/2, x_2^* = c/2$
- No discrimination: $x^* = (a+c)/2$
 - ➔ Output the same with and w/o discrimination
- NOTE: Not true in general with other types of demand

- If selling only in market 1 w/o discrimination

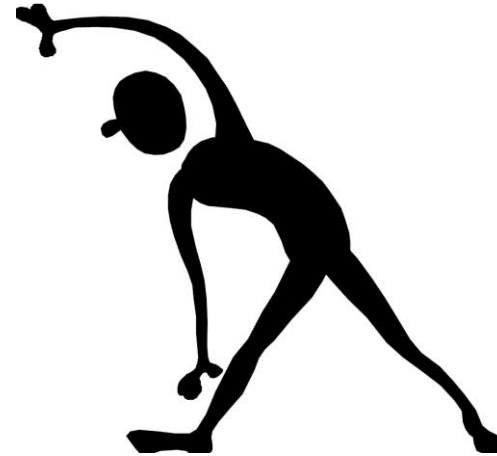
- Discrimination: $x_1^* = a/2, x_2^* = c/2$
- No discrimination: $x^* = a/2$
 - ➔ Output greater with discrimination



3rd degree price discrimination

Exercise

- Setup
 - Market 1 (high demand): $x_1 = 100 - p_1$
 - Market 2 (low demand): $x_2 = 100 - 2p_2$
 - $MC = 20$
- Questions:
 1. **If they can discriminate:**
 - a) What price will they charge in each market?
 - b) What will be the total output?
 2. **If the seller cannot discriminate:**
 - a) What price will they charge?
 - b) What will be the total output?
 - c) Will they sell in both markets?



Bundling

- **Bundling** = selling multiple products in fixed proportions as a package
- Examples:
 - TV channel bundles
 - TVs purchasing programmes from distributors
 - Microsoft Office
 - Set menus in restaurants
 - Machinery+maintenance packages



Bundling



- **Example**

- 2 TV channels: CNN, Eurosport
- 2 types of customers: serious guy, sports fan
- Without bundling, it is optimal to charge 5000 for each channel → profit = 10000
- But selling bundles for 6000 → profit = 12000
- Works well if **relative** WTP (across different products) varies across individuals.

	Serious guy	Sports fan
CNN	5000	1000
Eurosport	1000	5000

Tie-in sales as price discrimination

- **Tie-in sales** = selling products that require each other but not in fixed proportions
- Examples
 - Printers and cartridges
 - Cameras and lenses
- 2 types of customers:
 1. occasional printers (low WTP)
 2. frequent printers (high WTP)
- Each type of customers needs only 1 printer and has similar print quality requirements
→ 2nd degree discrimination not an option
- By requiring original cartridges, producer can
 - reduce the price of the printer and
 - extract the surplus through the cartridges
- **Intuition:** differential demand for printing used to discriminate between the customers



Please get in groups of 3

Quiz

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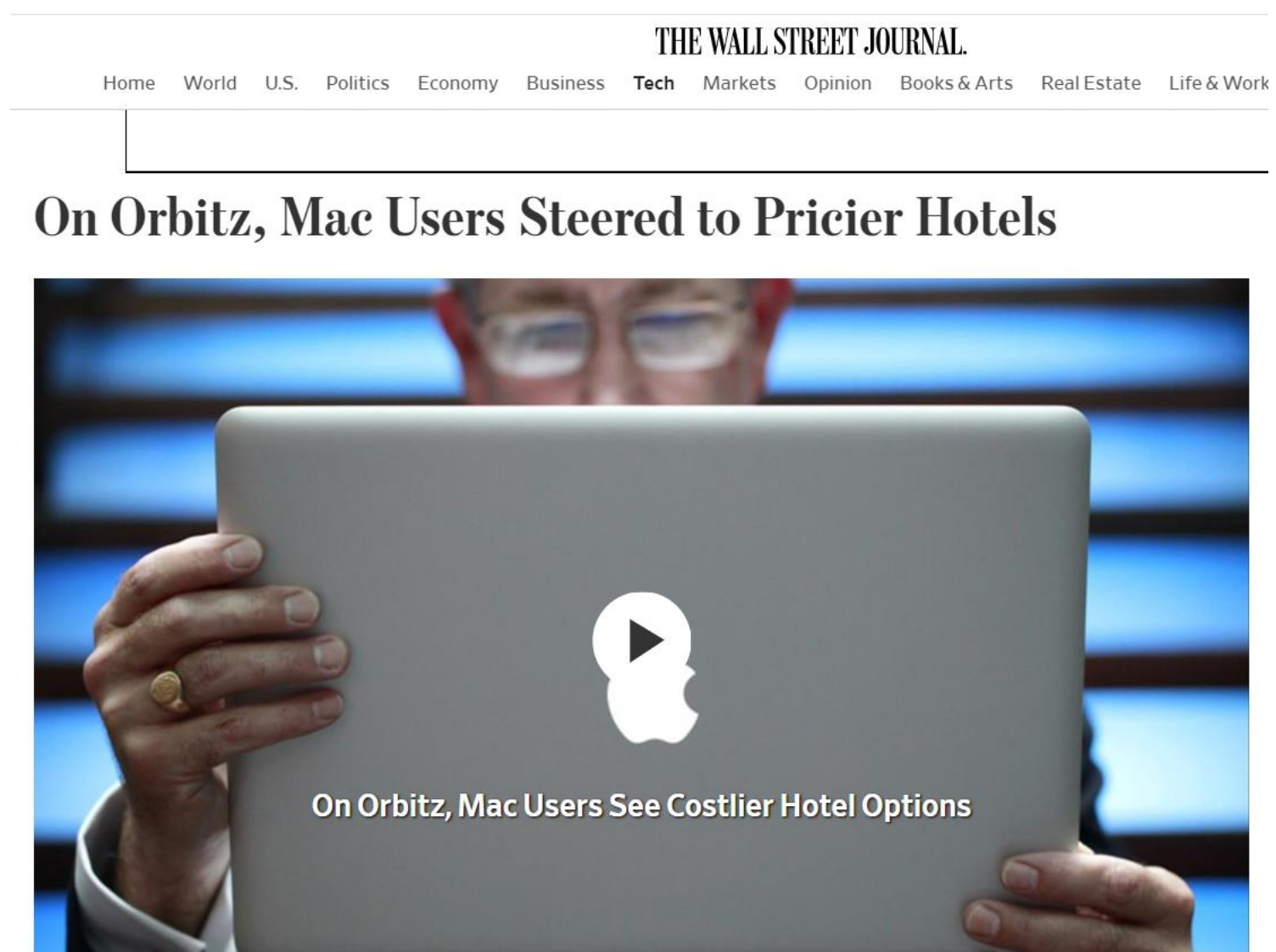
- What type(s) of price discrimination is it an example of?

Quiz



- Why do airlines charge much more for tickets purchased at the last minute? What type of price discrimination is it an example of?
- Why do Broadway theaters follow the opposite practice?

Quiz



- What type(s) of price discrimination is it an example of?

Quiz



- What type(s) of price discrimination is it an example of?